REDCap – Advanced Multi-Arm Assignment Instructions (Improved Structure)

1. User Account

Create or log into your REDCap account using:

https://redcap.uantwerpen.be/index.php
Use your **student email only**.

2. Project Creation & User Rights

1. One group member should create a **new longitudinal project**.
2. Add collaborators and the following REDCap accounts as *Project Users*:
   - **tafadzwa**
   - **redcap** (system admin)
3. Assign user roles:
   - **Data Managers**
   - **REDCap Admin**
4. Give **full access** to both roles.

3. Instruments to Build (Based on CRFs)

You must create the following instruments:

- **SELIG** – Screening & Enrollment Eligibility (page 4)
- **SEGE** – Screening/Enrollment Genital Exam (pages 5–6)
- **FSX** – Sexual History Follow-up (pages 7–8)
- **FGE** – Follow-up Genital Exam (pages 9–11)
- **MV** – Missed Visit (page 12)
- **Add ONE EXTRA Instrument: "Survey & Data Entry (SDE)" – Design your own form.**

Purpose:

- Practice advanced branching logic
- Apply form-level logic and field embedding
- Use Action Tags effectively

Mandatory SDE Features

Section A – Participant Locator Information

- Participant phone numbers
- GPS coordinates (dummy data allowed)
- Must use: **@CHARLIMIT**, **@LATITUDE**, **@LONGITUDE**

Section B – Contraception Knowledge Check

- Multiple-choice contraceptive methods
- Follow-up questions shown *only when relevant*

Example branching logic:
Show "How long have you used Injectables?" only if:

```
[contraception(2)] = '1'
```

Section C – Logic Validation Section

- Create calculated field:

```
total_condom_use = number_times_had_sex - number_times_no_condom
```

- Logic requirements:
  - If **total_condom_use < 0**, show warning field
  - Use: **@HIDECHOICE**, **@CALCTEXT**, Field Embedding

Section D – Eligibility Confirmation

- Mini version of SELIG logic
- Use branching logic to determine follow-up eligibility
- Final field:
  "Participant is eligible for follow-up: YES/NO (calculated)"

4. Apply Branching Logic Across all Forms

Use CRF rules to reproduce real clinical logic.

Examples (from SELIG, page 4):

- If **1a = No** → hide **1b**, jump to item 2
- If **2a = No** → mark as **Ineligible**
- Items **4–7** must all be "YES"
- For item 8:
  - If **never had sex** → skip to item 9
  - If **had sex AND lifetime partners >1** → ineligible

5. Add Three Arms (More Complex Design)

Convert the project into a **multi-arm longitudinal study**:

ARM 1 – Standard 12-Month Follow-Up

Mimics the original protocol.

ARM 2 – High-Frequency Follow-Up

Quarterly FGE + monthly FSX.

ARM 3 – Minimal Follow-Up

Only Screening, Month 6, and Month 12.

Each arm has its **own schedule** and **instrument allocation**.

6. Hypothetical Visit Structure (Required)

You MUST implement the following schedules:

ARM 1 – Standard Follow-Up (Original Structure)

Name: *"ARM 1 – Core Prospective Cohort"*

Event Name	Time Point	Instruments Assigned
Screening & Enrollment	Day 0	SELIG, SEGE, SDE
Month 1	+1 mo	FSX (repeat), FGE (repeat)
Month 3	+3 mo	FSX (repeat), FGE (repeat)
Month 6	+6 mo	FSX (repeat), FGE (repeat)
Month 9	+9 mo	FSX (repeat), FGE (repeat)
Month 12	+12 mo	FSX (repeat), FGE (repeat)
Missed Visit	Any	MV (repeat)

ARM 2 – High-Frequency Clinical Monitoring

Name: *"ARM 2 – Intensive Monitoring Arm"*

Event Name	Time Point	Instruments Assigned
Screening & Enrollment	Day 0	SELIG, SEGE, SDE
Month 1	+1 mo	FSX (repeat), FGE
Month 2	+2 mo	FSX ONLY (monthly repeat)
Month 3	+3 mo	FSX, FGE
Month 4	+4 mo	FSX ONLY
Month 6	+6 mo	FSX, FGE
Month 9	+9 mo	FSX ONLY
Month 12	+12 mo	FSX, FGE

Missed Visit	Any	MV (repeat)

Special Rules:

- FSX repeats **monthly**
- FGE only at **quarterly** visits
- Use branching logic to hide FGE items unless:
  [gen_exam_done] = '1'

ARM 3 – Minimal Follow-Up

Name: *"ARM 3 – Reduced Visit Arm"*

Event Name	Time Point	Instruments Assigned
Screening & Enrollment	Day 0	SELIG, SEGE, SDE
Month 6	+6 mo	FSX, minimal FGE
Month 12	+12 mo	FSX ONLY
Missed Visit	Any	MV (repeat)

Rules:

- FGE must hide all swab-related fields
- Only **external genital exam** allowed (SEGE page 5)

7. Configure Repeating Instruments

Configure:

- **FSX** → repeat in ALL arms
- **FGE** → repeat in ARM 1 & ARM 2
- **MV** → repeat in ALL arms
- **SDE** → optional repeat setting

8. Required Form Logic & Branching Logic

SELIG

- If **1a = No** → hide 1b, skip to 2
- If **8 = Yes AND lifetime_partners >1** → auto-ineligible
- If any of **9–11 = Yes** → "Not Eligible"

FGE

- If `[gen_exam_done] = '0'` → hide items 2–5
- If `[lesions] = '1'` → show HSV swab instruction

FSX

- If **no sex in past 3 months** → skip condom questions
- If **contraception = Yes** → show list of methods

SDE

Must include:

- `@CALCTEXT`
- `@HIDECHOICE`
- `@CHARLIMIT`
- Piping
- Validation for inconsistent sex/condom reporting

9. Move Project to Production

Verify:

- Branching logic functions correctly
- All fields are labeled
- Arms and events are mapped correctly
- Calculations validate correctly

Then submit for **Production Mode**.

10. Test Using REDCap Mobile App

- Install the app
- Generate Mobile App Code
- Enter test participants in each arm
- Upload offline → verify syncing

11. Multi-Centre Set-up: Data Access Groups (DAGs)

PISTIC is now a **multi-centre study** with three sites. Each site may see and enter only its own participants.

- Create three DAGs (Applications → Data Access Groups): **Dar es Salaam**, **Mwanza**, **Mbeya**
- Assign each group member to one DAG (one site per member)
- Keep **tafadzwa** and **redcap** without a DAG, so that they can see all sites
- Enter test participants in each DAG (all three arms)
- Log in as a site user and verify that only that site's records are visible
- Export the data with the `redcap_data_access_group` column

12. Group Task Requirements

Each group member must complete:

- **One full instrument**, AND

- **One advanced feature**
  (branching logic, action tags, field embedding, repeating events, multi-arm setup, Data Access Groups)

Deadline:

13 December 2026, 11:00